

Metaboenrich

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1 Algorithm for Undirected Metabolite-Enzyme Graphs

1.1 Graph Construction

An undirected graph $G = (V, E)$ is instantiated, where $V = V_m \cup V_r$ consists of metabolite nodes V_m and reaction nodes V_r . Edges E link metabolite nodes to the reaction nodes in which they participate.

1.2 Voltage Initialization

Each node $v \in V$ is assigned an initial voltage $\phi(v)$. Metabolite nodes present in the user’s differential abundance analysis (DAA) are initialized as

$$\phi(v) = |\log_2(\text{FC}(v))|, \quad v \in V_m^{\text{DAA}},$$

where $\text{FC}(v)$ is the fold change of metabolite v . All reaction nodes, and any metabolite nodes not present in the DAA, are initialized with

$$\phi(v) = 0.$$

1.3 Source and Target Node Definition

Source and target nodes are defined among the DAA metabolite nodes based on the direction and significance of change:

- **Source nodes** $S \subseteq V_m^{\text{DAA}}$: metabolites with a positive $\log_2$ fold change and a significant p -value.
- **Target nodes** $T \subseteq V_m^{\text{DAA}}$: metabolites with a negative $\log_2$ fold change and a significant p -value.

1.4 Current Flow Propagation

For each source–target pair $(s, t) \in S \times T$, current flow is computed over G using an approach that combines Kirchhoff’s current law and Ohm’s law, implemented via the graph Laplacian L . Treating s as a current source and t as a current sink, the node potentials x satisfy

$$Lx = b,$$

where b is the vector with $+1$ at s , -1 at t , and 0 elsewhere. Solving this system (e.g., via the Moore–Penrose pseudoinverse of L) yields the current flowing through each edge, and the voltage $\phi(v)$ of reaction nodes is updated based on the accumulated current passing through them across all source–target pairs.

1.5 Final Scoring

After current flow propagation, source, target, and reaction nodes are each analyzed for:

- **Conductivity score** – reflecting how efficiently current flows through the node across all source–target pairs.

Algorithm 1 Kirchhoff–Ohm Voltage Propagation

```
1: Construct Laplacian matrix  $L$  from  $G$ 
2: for each  $s \in S, t \in T$  do
3:   Set boundary vector  $b$ :  $b_s = 1, b_t = -1, b_v = 0$  otherwise
4:   Solve  $Lx = b$  for node potentials  $x$ 
5:   Compute edge currents from  $x$  using Ohm’s law
6:   Update voltage  $\phi(v)$  for reaction nodes  $v \in V_r$  using accumulated current
7: end for
```

- **Bottleneck score** – reflecting the extent to which a node constrains current flow (i.e., acts as a limiting point) across the network.

These two scores are combined to form each node’s final score, producing a ranked output over source, target, and reaction nodes.

2 Algorithm for Directed Metabolite-Enzyme Graphs

While the base algorithm treats G as undirected, many downstream questions – e.g., whether a signal can plausibly flow from a source metabolite to a target metabolite given known reaction stoichiometry – require directionality. Here we extend the graph and current-flow procedure to a directed setting in which edges encode the direction of substrate consumption and product formation, mediated by the enzyme(s) catalyzing each reaction.

2.1 Directed Graph Construction

We construct a directed graph $G' = (V, E')$ over the same node set $V = V_m \cup V_r$ used previously, where V_r now represents reaction/enzyme nodes. For each reaction $r \in V_r$ with substrate set $\text{Sub}(r) \subseteq V_m$ and product set $\text{Prod}(r) \subseteq V_m$, directed edges are added as

$$(m, r) \in E' \quad \forall m \in \text{Sub}(r), \quad (r, m') \in E' \quad \forall m' \in \text{Prod}(r).$$

For reversible reactions, both the forward edge set above and its reverse ($m' \rightarrow r \rightarrow m$) are included, optionally with a lower edge weight or capacity to reflect a directional preference (e.g., derived from ΔG or flux data) if available. Irreversible reactions retain only the forward direction.

2.2 Voltage and Boundary Conditions

Voltage initialization for V_m^{DAA} and source/target definitions (S, T) follow the same rules as in the undirected case (Sections 2.2–2.3). However, only source–target pairs $(s, t) \in S \times T$ for which a directed path exists from s to t in G' are considered biologically plausible and retained for current-flow propagation; pairs with no directed path are excluded from scoring or flagged separately.

2.3 Directed Current Flow

Kirchhoff’s and Ohm’s laws are applied to the directed graph using a variant of the graph Laplacian that respects edge direction. Let L' denote the directed (or “asymmetric”) Laplacian constructed from the incidence structure of E' , with edge conductances c_e assigned per the weighting scheme in Section 3.1 (uniform if no direction-specific weighting is available). For each retained pair (s, t) , node potentials x are obtained by solving

$$L'x = b,$$

with $b_s = 1, b_t = -1$, and $b_v = 0$ otherwise, subject to the constraint that current only flows along the direction permitted by E' . Reaction node voltages $\phi(v)$, $v \in V_r$, are updated by accumulating the resulting directed current across all retained source–target pairs.

Algorithm 2 Directed Kirchhoff–Ohm Voltage Propagation

```
1: Construct directed graph  $G'$  from reaction substrate/product sets, including reversible edges where applicable
2: Construct directed Laplacian  $L'$  with edge conductances  $c_e$ 
3: for each  $s \in S, t \in T$  do
4:   if a directed path exists from  $s$  to  $t$  in  $G'$  then
5:     Set boundary vector  $b$ :  $b_s = 1, b_t = -1, b_v = 0$  otherwise
6:     Solve  $L'x = b$  for node potentials  $x$ , respecting edge direction
7:     Compute directed edge currents from  $x$  using Ohm's law
8:     Update voltage  $\phi(v)$  for reaction nodes  $v \in V_r$  using accumulated directed current
9:   else
10:    Exclude  $(s, t)$  from scoring (no plausible directed path)
11:   end if
12: end for
```

2.4 Scoring Adjustments

Conductivity and bottleneck scores are computed as in Section 2.5, restricted to the retained directed source–target pairs. Because directionality removes biologically implausible paths, we expect bottleneck nodes identified under this scheme to more precisely localize points of pathway constraint (e.g., a single irreversible, rate-limiting enzyme) rather than artifacts of the undirected topology.